gpeyt1-22 · Infinite slope with pore pressure

GEO PE · Long Calculation Problems I — Topic 5: Earth Structures, Ground Improvement & Pavements

Problem

A long natural hillside of residual silty sand slopes at β = 18°. A shallow slip surface parallel to the slope is suspected at z = 8 ft (vertical depth). γ = 122 pcf, c′ = 0, φ′ = 30°. During wet season a piezometer on the slip surface reads u = 280 psf. The road at the toe carries 2,000 vehicles/day.

Most nearly, the factor of safety during the wet season is:

A. 1.78
B. 1.27
C. 1.21
D. 0.68

Given equations / data

> $r_u = u/(\gamma z)$; $F = [1 - r_u(1 + \tan^2\beta)] \tan\phi'/\tan\beta$

Worked solution

1. $r_u = 280/(122 \times 8) = 0.287$
2. $1 + \tan^2 18 = 1.1056$
3. $F = [1 - 0.287 \times 1.1056] \times \tan 30/\tan 18 = 0.6828 \times 1.777 = 1.21$

Final answer

1.21

Why the other values are wrong

- **1.78** — dry case (ignored u)
- **1.27** — omitted the (1 + tan²β) term
- **0.68** — used γw instead of γ in ru

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.6.4 Infinite Slope, p. 105

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=SXA_TC3XEwk (no creator values copied).

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